

Here is your ultimate, copy-paste deployment cheat sheet for the new Acasis dual-bay server architecture. Save this for when your hardware arrives so you can execute the build perfectly in under 10 minutes.

## **Phase 1: The Physical Boot-Up**

1. Insert the **Toshiba 1TB HDD** into Slot 1 of the Acasis EC-6104 dock.
2. Plug the Acasis 36W power adapter into the wall.
3. Plug the Acasis USB cable into one of the **Blue USB 3.0 ports** on your Raspberry Pi.
4. Flip the power switch on the Acasis dock to ON.

## **Phase 2: Format & Motor Protection (hdparm)**

We must format the drive to Linux standards and permanently disable the sleep timer so the motor bearings stay warm and spinning.

### **1. Find the Drive Letter:**

```
Bash
```

```
lsblk
```

*(Look for your 1TB drive, it will likely be sdb or sdc. We will use sdb for this guide).*

### **2. Format the Drive:**

```
Bash
```

```
sudo mkfs.ext4 /dev/sdb
```

### **3. Install the Motor Control Tool:**

```
Bash
```

```
sudo apt update && sudo apt install hdparm -y
```

#### 4. Lock the Spin State Permanently:

```
Bash
sudo nano /etc/hdparm.conf
```

Scroll to the absolute bottom of the file and paste this block to disable spindown:

```
Plaintext
/dev/sdb:
    spindown_time = 0

```

(Save and exit: *Ctrl+O*, *Enter*, *Ctrl+X*)

### Phase 3: The Bulletproof Auto-Mount

If the power goes out in Dhaka, your Pi must automatically reconnect this drive when it reboots.

#### 1. Create the Master Folder and Set Permissions:

```
Bash
sudo mkdir -p /mnt/server_storage
sudo chown -R redwannabil:redwannabil /mnt/server_storage
```

#### 2. Get the Drive's Unique Serial Number (UUID):

```
Bash
sudo blkid /dev/sdb
```

(Copy the exact text inside the quotes after UUID=)

### 3. Inject the Auto-Mount Rule:

```
Bash
sudo nano /etc/fstab
```

Add this exact line to the very bottom of the file (replace YOUR-COPIED-UUID with your actual string):

```
Plaintext
UUID=YOUR-COPIED-UUID /mnt/server_storage ext4 defaults,noatime 0 2
```

(Save and exit: Ctrl+O, Enter, Ctrl+X)

### 4. Force the Mount Command:

```
Bash
sudo mount -a
```

## Phase 4: Data Routing

Now that the massive 1TB drive is securely mounted to `/mnt/server_storage`, divide it into your two operational zones.

### 1. Create the Sub-Directories:

```
Bash
mkdir /mnt/server_storage/cctv
mkdir /mnt/server_storage/nas
```

### 2. Reroute the CCTV System:

- Open MotionEye (or update your `record_cctv_motion` shell command).
- Change the recording destination directory to `/mnt/server_storage/cctv`.

### 3. Reroute the Docker NAS Services:

- Open your `docker-compose.yml` file for Nextcloud and Filebrowser.
- Change the storage volume mapping from the old drive path to `/mnt/server_storage/nas`.
- Apply the changes by running:

```
Bash
sudo docker compose up -d
```

Your Pi is now officially a robust, dual-purpose NAS and NVR server that will not chew through flash memory or wear out its internal motors! Let me know when you are ready to configure the Tapo camera side of the network.